INCH - POUND

MIL-PRF-27/289B
W/AMENDMENT 2
7 January 2022
SUPERSEDING
MIL-PRF-27/289B
AMENDMENT 1
10 May 2017

PERFORMANCE SPECIFICATION SHEET

TRANSFORMERS AND INDUCTORS (AUDIO, POWER AND HIGH-POWER PULSE) INDUCTORS, POWER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the transformer described herein shall consist of this document and the latest issue of specification MIL-PRF-27.

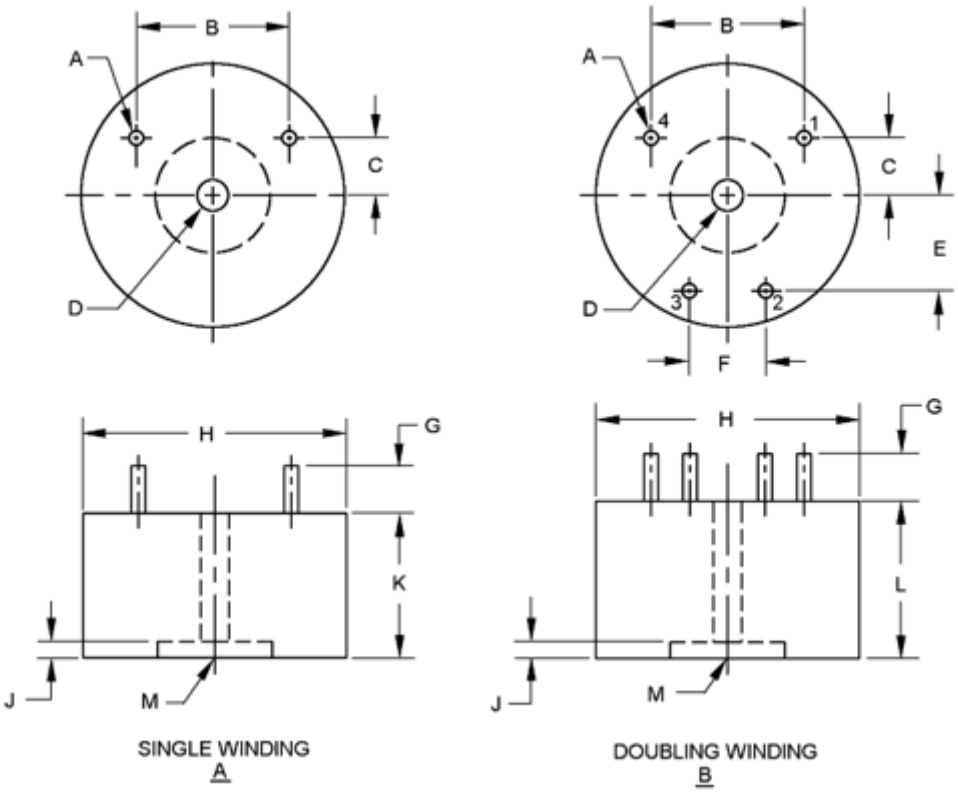
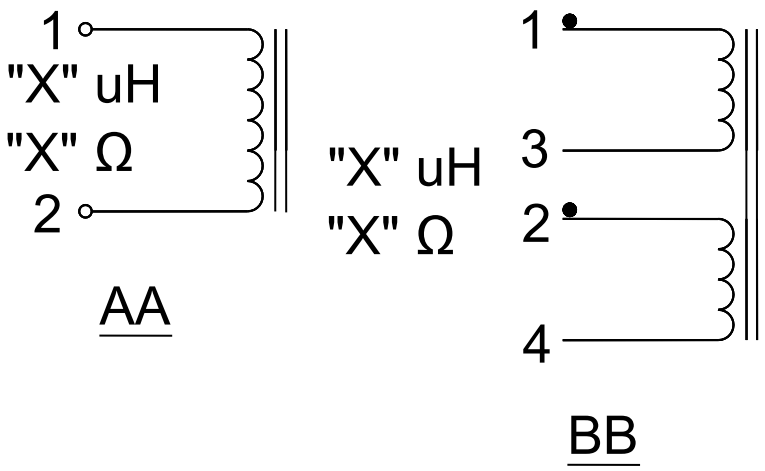


FIGURE 1. Dimensions and configurations.

AMSC N/A
FSC 5950

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WV – 357V
(SEE TABLE I FOR VALUES OF “X”)
CIRCUIT DISGRAM AND MARKING

Dim	Inches	mm	Dim	Inches	mm
A	.07 ±.005	1.78 ±0.13	G	.25 ±.030	6.35 ±0.76
B	.80 ±.010	20.32 ±0.25	H	1.37 ±.030	34.80 ±0.76
C	.30 ±.010	7.62 ±0.25	J	.08 ±.015	2.03 ±0.38
D	.15 ±.010	3.81 ±0.25	K	.75 ±.063	19.05 ±1.60
E	.50 ±.010	12.70 ±0.25	L	.81 ±.063	20.57 ±1.60
F	.40 ±.010	10.16 ±0.25	M	.28 ±.015	7.11 ±0.38

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking shall be on the side and on top.
4. Electrical values shall be marked as specified in table I, as applicable.
5. Circuit diagram BB, for series connection join terminals 2 and 3 for parallel connection join terminals 1 and 2, 3 and 4.

FIGURE 1. Dimensions and configurations – Continued.

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REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings: See table I.

TABLE I. Electrical ratings.

Dash no. 1/	Inductance (µh) +15 percent -5 percent 2/ 3/	DC current (A) max 2/ 4/	DC resistance (ohms) max 2/	Working voltage (peak) volts 5/	PIN arrangement (see figure 1)	Circuit Diagram (see figure 1)	Voltage
01	10,000	1.1	2.1	357	A	AA	1 volt at 10 kHz
02	6,400	1.35	1.4	357	A	AA	1 volt at 10 kHz
03	2,500	2.2	.55	357	A	AA	1 volt at 10 kHz
04	1,600	2.8	.34	357	A	AA	1 volt at 10 kHz
05	1,000	3.5	.21	357	A	AA	1 volt at 10 kHz
06	640 / 160	4.5 / 9	.13 / .033	357	B	BB	1 volt at 10 kHz
07	400 / 100	7.5 / 10.8	.088 / .022	357	B	BB	1 volt at 10 kHz
08	240 / 60	6.8 / 13.6	.056 / .014	357	B	BB	1 volt at 10 kHz

1/ Qualification test and approval to M27/289-01, shall be sufficient to grant qualification approval to M27/289-02 through -08.
2/ Where electrical values are written one above the other they indicate the values from series or parallel connections respectively $\frac{\text{(series)}}{\text{parallel}}$
3/ The inductance is measured with 0 A dc applied at the specified voltage and frequency.
4/ At rated dc current, inductance will exceed 80 of rated inductance.
5/ Between windings, 50 volts peak.

Design and construction:

Dimensions and configuration: See figure 1.

Duty cycle: Continuous.

Case: Epoxy.

Terminals: Printed circuit (tinned brass).

Weight: 3 ounces.

Altitude: 75,000 feet.

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Operating temperature range: -55°C to +130°C.

Terminal strength: MIL-STD-202-211, test condition A, 2 pounds.

Dielectric withstanding voltage:

At sea level: 1,000 Vrms.

At reduce barometric pressure: 500 Vrms.

Vibration (high frequency): MIL-STD-202-204.

Temperature rise: +45°C with 1 volt rms, 10 kHz, current as specified in table I at an ambient temperature of +85°C maximum.

Marking location: See figure 1.

Part or Identification Number (PIN): M27/289-(dash number from table I).

Referenced documents. In addition to MIL-PRF-27, this document references the following:

MIL-STD-202-204 MIL-STD-202-211

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5950-2022-011)

Review activities:
Army - AR, MI
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